

Monte Carlo Project Risk Analysis

Generated by uber-polya

2026-02-26

Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Simulation / Monte Carlo

Universe

- Statistics: $\{\text{mean_duration, std_duration, median_duration, ci_95_low, ci_95_high}\}$
- Risk Metrics: $\{\text{prob_finish_by_deadline, var95, cvar95, deadline}\}$
- Convergence: $\{\text{sample_sizes, running_means}\}$
- Analytical Crosscheck: $\{\text{analytical_mean, mc_mean, difference}\}$

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

A software project has 5 phases, each with uncertain duration modeled as a triangular distribution (min, mode, max in days). Phases 1-3 are sequential; phases 4 and 5 run in parallel after phase 3 completes.

Phase estimates (days):

Phase	Min	Mode	Max	Dependencies
Phase 1	5	8	14	None
Phase 2	3	5	10	Phase 1
Phase 3	7	10	18	Phase 2
Phase 4	4	6	12	Phase 3
Phase 5	6	9	15	Phase 3

Critical path: Phase 1 + Phase 2 + Phase 3 + max(Phase 4, Phase 5)

Questions:

- What is the expected total project duration (mean and 95% CI)?
- What is the probability of finishing within 50 days?
- What is the Value at Risk at 95% (worst-case duration at 95th percentile)?
- Which phase is the critical risk driver?

Real-World Mapping

Real-World Concept

problem input

Mathematical Object

instance parameters

Constraints

- (1) Monte Carlo simulation**: Estimate distributional properties of complex systems by repeated sampling
- (2) Triangular distribution**: Three-parameter distribution (min, mode, max) for expert elicitation
- (3) Critical path**: The longest dependency chain determines total project duration
- (4) Value at Risk (VaR)**: The duration threshold exceeded only $(1 - \text{confidence})\%$ of the time
- (5) Conditional Value at Risk (CVaR)**: Expected duration in the worst-case tail beyond VaR
- (6) Convergence analysis**: Demonstrates that Monte Carlo estimates stabilize as sample size increases
- (7) Sensitivity analysis**: Correlation between individual phase durations and total duration
- (8) Confidence intervals**: Quantify estimation uncertainty in the Monte Carlo mean via the Central Limit Theorem

Approach

Suggested approach Monte Carlo simulation with triangular distributions

Complexity class:

Available tools: Monte Carlo simulation with triangular distributions

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	Monte Carlo simulation with triangular distributions
Time:	0.0459s

Solution Details

Solution computed successfully.

Verification

✓ Mean Reasonable	Passed
✓ Std Positive	Passed
✓ Ci Contains Mean	Passed
✓ Var95 Gt Mean	Passed
✓ Cvar95 Ge Var95	Passed
✓ Prob Deadline Between 0 1	Passed
✓ Convergence Stabilizes	Passed
✓ Analytical Close	Passed
✓ Sensitivity Sums Reasonable	Passed
✓ Simulation Count Correct	Passed
✓ All Passed	Passed

3. Interpretation

The Question

A software project has 5 phases, each with uncertain duration modeled as a triangular distribution (min, mode, max in days).

The Answer

A feasible solution was found.

What This Means

- Mean duration: ~36 days
- 95% CI for mean: narrow interval around the mean (via CLT)
- P(finish $\leq$ 50 days): high probability ($>90\%$)
- VaR95: 95th percentile duration (~43-44 days)
- CVaR95: conditional tail expectation above VaR95
- Critical risk driver: Phase 3 or Phase 5 (highest correlation with total duration)
- Convergence: running mean stabilizes well before 100,000 simulations

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution